

ASCM CORE PLATFORM

ASCM Core Platform

Autonomous Software Engineering & Agentic Coding Machine

ASCM Enterprise Commercial & Architectural Brief | Author: ASCM Core Squad | Release: Production v1.0

Why Current AI Coding Assistants Fail in Real Engineering

Copilot, Cursor, and ChatGPT work on single files or single repos, ignoring distributed system reality.

Single-Repo Myopia • Real systems span 5 to 50 microservices, SDKs, and frontend repos. • Changing an API parameter silently breaks 3 downstream consumer services. • Single-repo AI assistants have zero cross-repository dependency awareness.	Confirmation Bias Trap • When the same LLM writes and tests code, it confirms its own hallucinations. • Mock tests pass locally while production microservices crash under runtime loads. • Zero adversarial rigor in existing AI code generation workflows.	Siloed GTM Disconnect • Engineering ships features, but sales and marketing never hear about them. • Zero automated connection between git commits and customer pipeline generation. • Weeks of delay translating technical accomplishments into business growth.
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ASCM's 4 Core Pillars of Autonomous Engineering

A complete, self-healing software factory backed by adversarial multi-agent orchestration.

1. Multi-Repo Dependency Graph

- Builds complete multi-repo ASTs and dependency graphs.
- Simulates breaking changes across contracts before generating code.
- Ensures atomic cross-repository commits.

2. Adversarial Critic Verification

- Two-phase verification: Independent critic agents actively try to break code.
- Runs mathematical proofs, fuzzing, and contract validation before commit.
- Eliminates confirmation bias completely.

3. Domain Adaptation & GTM Loop

- LoRA fine-tuning and INT4 GGUF quantization for edge deployment.
- Native Sales & Marketing agents directly tied to git commits.
- Instant zero-setup multi-channel distribution and pipeline closing.

What ASCM Built: Real-World Production Systems

Not a toy demo. ASCM generates, verifies, and deploys mission-critical enterprise applications.

Mutual Funds SLM Copilot • Fine-tuned edge financial advisor running in 2.2GB RAM. • Deterministic math grounding (CAGR, Sharpe, Alpha). • Zero hallucination guarantee under SEBI/SEC guardrails. • Live 24/7 web UI & AMFI NAV sync.	PayPulse Sentinel Gateway • High-throughput multi-repo payment & crypto escrow system. • Cross-repo contract verification between gateway and settlement. • Self-healing automated rollbacks under failure. • 100% test coverage under chaos tests.	CAD/CAM Toolpath Engine • Physics-grounded manufacturing CNC code generator. • Feed rate, spindle speed, and tool deflection safety proofs. • Validated against industrial tolerances. • Zero-shot G-code generation.
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Enterprise Economics: 10x Velocity, Zero Breaking Changes

Transforming engineering departments from maintenance cost centers into high-speed innovation hubs.

78% Fewer Outages • Eliminates cross-service breaking changes. • Catches schema drift before PR merge. • Saves millions in production downtime.	5x Shipping Speed • Autonomous multi-repo PRs generated in minutes. • Developers review and approve, rather than write boilerplate. • Massive time-to-market advantage.	Zero Local AI Bill • Quantized INT4 SLMs run on developer laptops. • Zero API cost per developer seat for local copilot tasks. • Complete enterprise data privacy.	Built-In GTM Engine • Marketing and Sales run autonomously alongside engineering. • Instant pipeline creation upon feature release. • The complete founder & enterprise operating system.
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